NEXORA System Analysis and Design

Undergraduate Software Engineering Project - Professor Farrokhian

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Three phases, from requirements to analysis and design models

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Purpose and Basis of the Report

Subject

This report describes Nexora's requirements and design from the perspectives of users, administrators, and software components. It defines system scope, operational flows, and data relationships for project evaluation and development.

Basis of Description

Existing capabilities are described using the project's code and documentation. Quality requirements and acceptance criteria are evaluation measures. They do not certify that tests have run or guarantee quality.

Guide to the Report

Section	Contents
Phase One	Problem definition, scope, stakeholders, user and system requirements, and quality requirements
Phase Two	System context, use cases, operational scenarios, and activity diagrams
Phase Three	Sequence diagrams, domain classes, data relationships, and implementation architecture
Evaluation and Development	Requirements traceability, acceptance criteria, proposed capabilities, and references

How to Read the Identifiers

UR denotes a user requirement, FR a functional system requirement, and NFR a nonfunctional requirement. UC denotes a use case and TC a test scenario. The traceability table links these identifiers.

Key Terms

A knowledge set defines a thematic scope for documents and permissions. A chunk is a smaller segment of document text indexed for retrieval. Retrieval-augmented generation (RAG) first retrieves relevant document segments and then supplies them and the question to the language model. A citation identifies the source used in an answer and does not guarantee that every sentence is correct.

The diagrams use English class and component names to make their correspondence with the code clear. The scenario explanations and figure guides in the original report are written in Persian.

Phase One: Problem Definition and Scope

Problem

Organizational documents are often scattered across multiple files and sources. To answer a question, a user must find the right document, read the relevant material, and identify the supporting source. Nexora organizes documents into thematic sets and lets users ask questions within their authorized sources.

Scope of the Current Version

The scope covers sign-in and sessions, member management, knowledge sets, document upload and processing, search, conversations, specialized assistants, source connections, conversation sharing, feedback, and administrative analytics and evaluation tools. The upload path supports PDF and TXT documents and JPG, PNG, and TIFF images. Extraction quality depends on the content type and OCR configuration.

System Responsibility Boundary

The system displays language model output with sources. Sensitive content still requires expert review. Operators are responsible for the accuracy of input files, permission to use the content, and configuration of external source credentials. Data from another organization or an unauthorized set falls outside a user's permitted answer scope.

Organization Boundary

The data model and API use an organization identifier, and the sign-in API accepts an organization name. The current sign-in form sends the value default. Organization selection in the sign-in interface and full organization lifecycle management are therefore not treated as completed capabilities in this report.

Stakeholders and Roles

Role	Needs and responsibilities
User	Ask questions of authorized sources, view citations, manage their own conversations, and submit feedback.
Organization administrator	Manage the organization's members and permissions, sets, documents, assistants, and connectors.
System operator	Deploy the API and workers, configure external services, monitor operations, and make backups.
External provider	Handle model or OCR requests or supply content for synchronization.

Role and Set Permission

A user's general role is distinct from their permission for a specific knowledge set. The set permission model stores view, edit, and manage levels. The API must decide whether an operation is allowed for the relevant role and permission. Hiding the option in the UI alone is insufficient.

Operating Prerequisites

The database, file storage, and Qdrant must be reachable by the relevant components. The Document Worker processes queued work, and the Connector Scheduler handles scheduled jobs. External models and OCR require configuration and network access. A service outage must produce a recognizable error state.

Functional Requirements 1

Authentication and Sessions

UR-01 Users shall be able to sign in with an active account and end their session.

FR-01 The system shall validate username and password within the supplied organization and reject inactive accounts.

FR-02 The system shall maintain a valid session using an HttpOnly cookie and reject requests without a valid session.

FR-03 The system shall revoke the associated session on sign-out and control failed sign-in attempts through rate limiting.

Member and Access Management

UR-02 Administrators shall be able to determine membership and access to organizational knowledge.

FR-04 The system shall create and edit users, roles, and active status within the administrator's authority.

FR-05 The system shall store a user's permission separately for each set and check it for relevant operations.

FR-06 The system shall exclude unauthorized sets and sources from the user's lists and retrieval results.

Knowledge Sets

UR-03 Administrators shall be able to organize documents in thematic sets.

FR-07 The system shall store set names, descriptions, and their relationships to documents.

FR-08 The system shall check access when creating, editing, or deleting sets.

FR-09 The system shall preserve the scope of one or more selected sets in conversation requests.

Functional Requirements 2

Document Upload

UR-04 Authorized users shall be able to add a valid file to a set and see the result.

FR-10 The system shall validate file type, size, and content before acceptance. Renaming the extension shall not bypass validation.

FR-11 The system shall store accepted files and register the corresponding document and processing job.

FR-12 The system shall display success or failure for each file beside the operation and keep other files' states independent.

Processing and Indexing

UR-05 Authorized users shall be able to see processing progress and why a document failed.

FR-13 The worker shall perform text extraction, OCR when needed, chunking, and indexing.

FR-14 The system shall record stage, progress, and error reason. A failed document shall not be marked ready to answer questions.

FR-15 The system shall offer pause only at pausable stages and accept resume or retry only in a valid state.

Source Search

UR-06 Users shall be able to find relevant material in authorized sources.

FR-16 The system shall determine authorized document IDs before vector search.

FR-17 The system shall prepare relevant chunks with source information for answer generation.

FR-18 The system shall distinguish unavailable sources and lack of relevant results from provider errors.

Functional Requirements 3

Conversations and Source-Grounded Answers

UR-07 Users shall be able to ask questions, read answers and sources, and retain history.

FR-19 The system shall prepare the question, necessary history, assistant settings, and retrieved content for the model.

FR-20 The system shall store and display the answer, sources, and grounding status in the associated message.

FR-21 The system shall identify truncation caused by generation limits and continue an answer without changing its access scope.

Specialized Assistant

UR-08 Administrators shall be able to define an assistant with specific instructions and sources.

FR-22 The system shall store the assistant's name, description, instructions, model, and active status.

FR-23 The system shall record associated sets and check them against the user's access when the assistant is used.

FR-24 The system shall prevent display or use of an assistant outside the organization and authorized scope.

Source Connections

UR-09 Administrators shall be able to import external content into a set and track synchronization.

FR-25 The system shall retrieve source content through the adapter for the connector type.

FR-26 The system shall track source items and content changes by external ID and content hash.

FR-27 The system shall store synchronization state, execution time, and error reason, and control concurrency through a lease.

Functional Requirements 4

Sharing and Feedback

UR-10 Users shall be able to share conversations in a controlled manner and evaluate answers.

FR-28 The system shall create a read-only conversation snapshot with a revocable link.

FR-29 The system shall enforce share revocation. A link shall not grant unrestricted search of organizational knowledge.

FR-30 The system shall associate feedback with the relevant answer and prevent submission for unauthorized answers.

Research and Evaluation

UR-11 Administrators shall be able to review research flows and system evaluation data.

FR-31 The application service shall coordinate research retrieval, generation, and citation construction.

FR-32 The system shall provide model usage data and evaluation items to the relevant administrative tool.

FR-33 The system shall control access to administrative tools independently of conversation access.

Interface and Operation Feedback

UR-12 Users shall be able to identify operation status without losing the page context.

FR-34 The system shall show an appropriate waiting or skeleton state during initial loading or data retrieval.

FR-35 The system shall show success messages inline near the relevant action.

FR-36 The system shall provide understandable errors, a way to correct input, and an appropriate retry action.

Product Nonfunctional Requirements

NFR-01 Security and Isolation

Every read or write shall check organization, ownership, and the relevant permission. Acceptance requires rejection of direct requests for unauthorized resource IDs and exclusion of those resources from retrieval. Verify this through negative tests.

NFR-02 Data Integrity

Related writes shall have a defined transaction boundary. SQL writes and vector indexing do not share a transaction. The outbox shall make indexing traceable and retryable. Acceptance requires recovery of incomplete work after an outage without falsely marking the document ready.

NFR-03 Usability

The Persian interface shall be right-to-left, display waiting states, and show operation errors in context. Acceptance requires completing sign-in, upload, and conversation with the keyboard and checking the layout at different sizes.

NFR-04 Performance

File receipt, indexing, and generation times shall be measured separately. This version presents no approved numerical target or load-test result. Capacity assessment requires a defined data volume, concurrent user count, and acceptable threshold.

NFR-05 Recoverability

Worker or external service failure shall leave the job state identifiable. Leases, heartbeats, and retries shall prevent unowned execution. PostgreSQL and file backups shall be restored in a test environment.

Organizational and External Requirements

NFR-06 Maintainability

HTTP routes shall validate requests and transform responses. Services shall coordinate operations, and repositories shall perform queries. Application service errors shall be independent of HTTP and external provider names. Verify with unit tests and dependency review.

NFR-07 Observability

Logs shall make it possible to diagnose failed requests, processing jobs, and connectors. Public logs shall exclude secrets such as passwords and credentials. Readiness shall reflect required dependencies and workers.

NFR-08 Deployment

The API, worker, and scheduler shall use compatible settings and connect to shared storage. Schema changes shall use Alembic migrations. Docker Compose is the current deployment method and does not itself guarantee high availability.

NFR-09 External Dependencies

Timeouts and failures for model calls, OCR, and content fetching shall be bounded and distinguishable. An external outage shall never produce false success. Provider selection and content submission shall follow the organization's data policy.

NFR-10 Accessibility and Compatibility

Key controls shall have recognizable labels, visible focus, and appropriate keyboard behavior. Browser compatibility and full accessibility compliance require dedicated testing. This report makes no claim of complete verification.

Phase Two: System Context Model

The context model treats Nexora as a single unit and shows its interactions with users and external services. The database and file store are internal implementation components and are not user actors in this view.

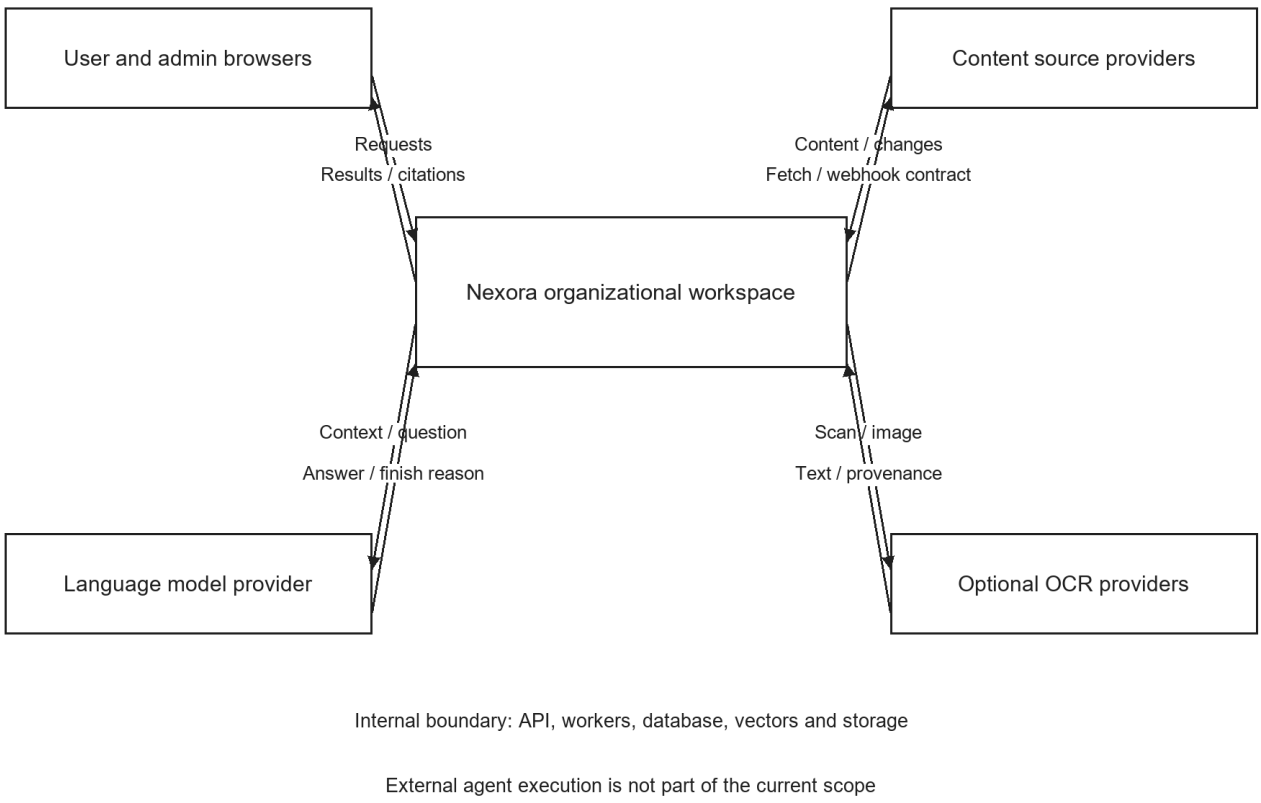


Figure 1. Nexora system context model

The user's browser sends questions and feedback. Administrators perform authorized management operations through the same interface. External sources supply content for ingestion into a knowledge set. Language model and OCR services return processing results. Data transmission to these services shall follow the operator's settings and policy.

Use Case Diagram

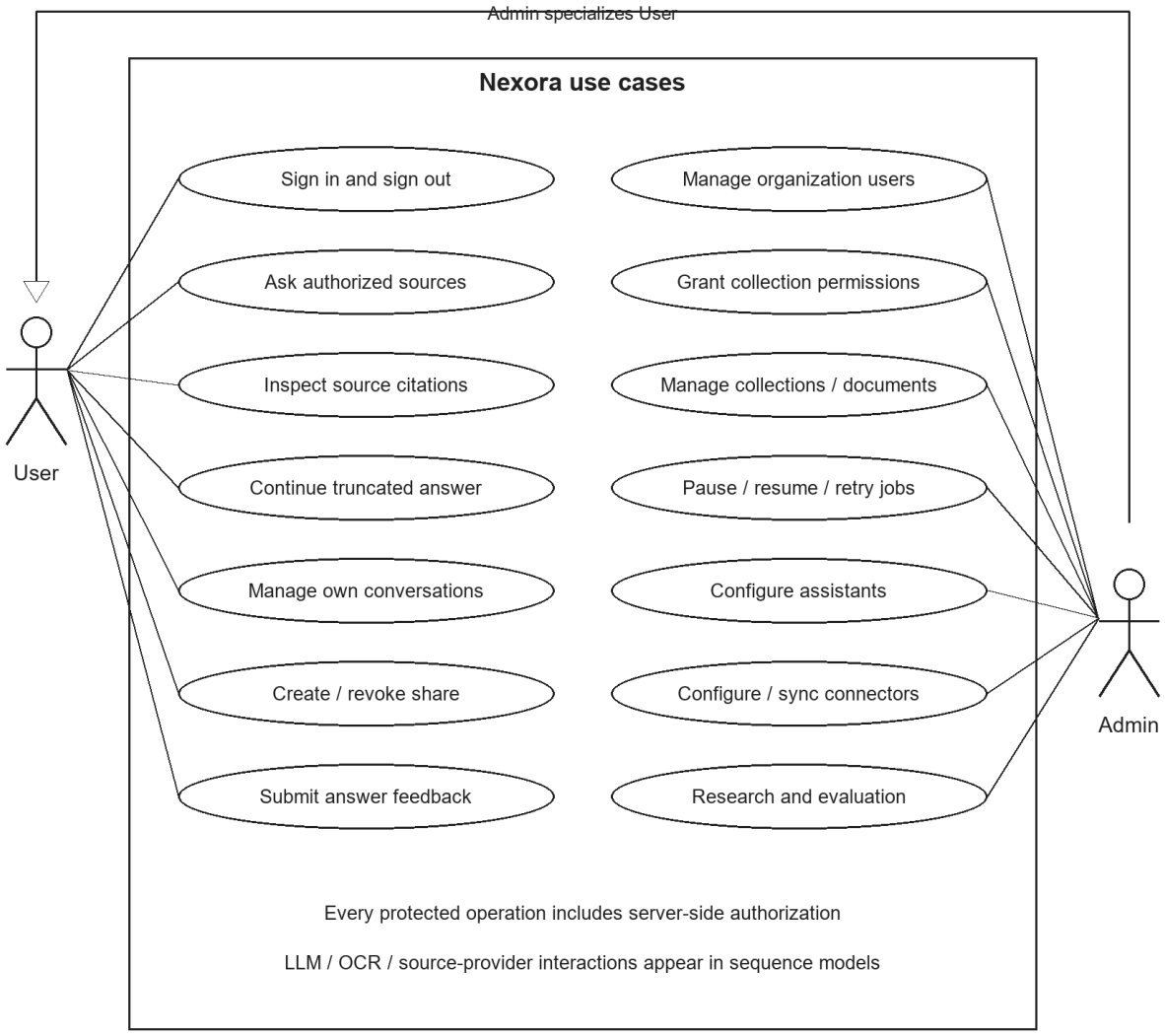


Figure 2. Main use cases within the Nexora boundary

Administrators can also perform ordinary user operations. Shared associations are omitted for legibility. Lines associate actors with operations and do not specify execution order. The “Ask authorized sources” use case always requires an access check, even when the user selects an assistant.

Main Use Cases

UC-01 Sign-in

The actor is a user or administrator. An active account and valid organization are required. The user submits credentials. The system checks rate limits, organization, account, and password and creates a session on success. On failure, it creates no new session and returns an appropriate message. A successful session permits authorized operations.

UC-02 Upload and Processing

The actor has the required permission for the set. A file and set are selected. The API checks permission and file validity and registers a processing job. The worker extracts and indexes the content. Invalid files are rejected before processing. Processing failures set an error state, and the document becomes retrievable only after the required work completes.

UC-03 Question and Answer

The actor has a valid session. The user enters a question and selects a knowledge scope or assistant. The system identifies authorized sources, retrieves relevant content, and generates an answer. It stores the answer and citations. Missing sources, interrupted generation, and provider errors must have distinct states.

UC-04 Assign Access

The actor is an organization administrator. The administrator selects a user and set and assigns or removes a permission level. The system checks that they belong to the same organization, verifies the administrator's authority, and records the change. Subsequent requests must apply the new permission. Prior UI access cannot replace the API check.

Additional Use Cases

UC-05 Define an Assistant

The administrator records the name, instructions, status, and associated sets. The system checks uniqueness and validity within the organization. The user can use only the intersection of the assistant's scope and their authorized sources. The current specialized assistant does not autonomously execute external tools.

UC-06 Synchronize a Connector

The administrator defines a connector for an authorized set. After acquiring a lease, the coordinator calls the provider adapter. New or changed items enter storage and indexing. Fetching or indexing failures have a traceable status and reason. Repeated runs must use external IDs to avoid duplicate versions.

UC-07 Share a Conversation

The authorized conversation owner requests a share. The system creates a read-only snapshot and link. A link holder can view the published content. The link does not grant unrestricted access to documents. The owner can revoke the share, after which requests to the link must be denied.

UC-08 Continue an Answer

The user requests continuation for a message marked as truncated. The system checks the conversation and current authorized sources and continues generation using relevant context. Permission changes between requests must be applied. A continuation error must not falsely mark an answer as complete.

Sign-in Activity

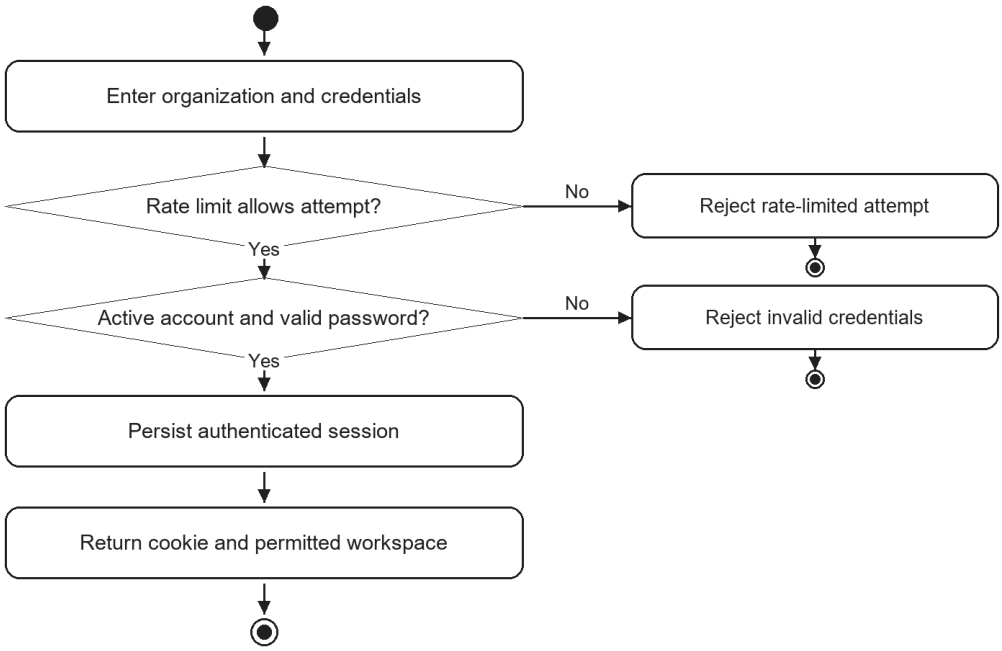


Figure 3. Sign-in activity diagram

The rejection branch creates no session. The organization name is part of API validation. The current interface still sends the default value.

Document Processing Activity

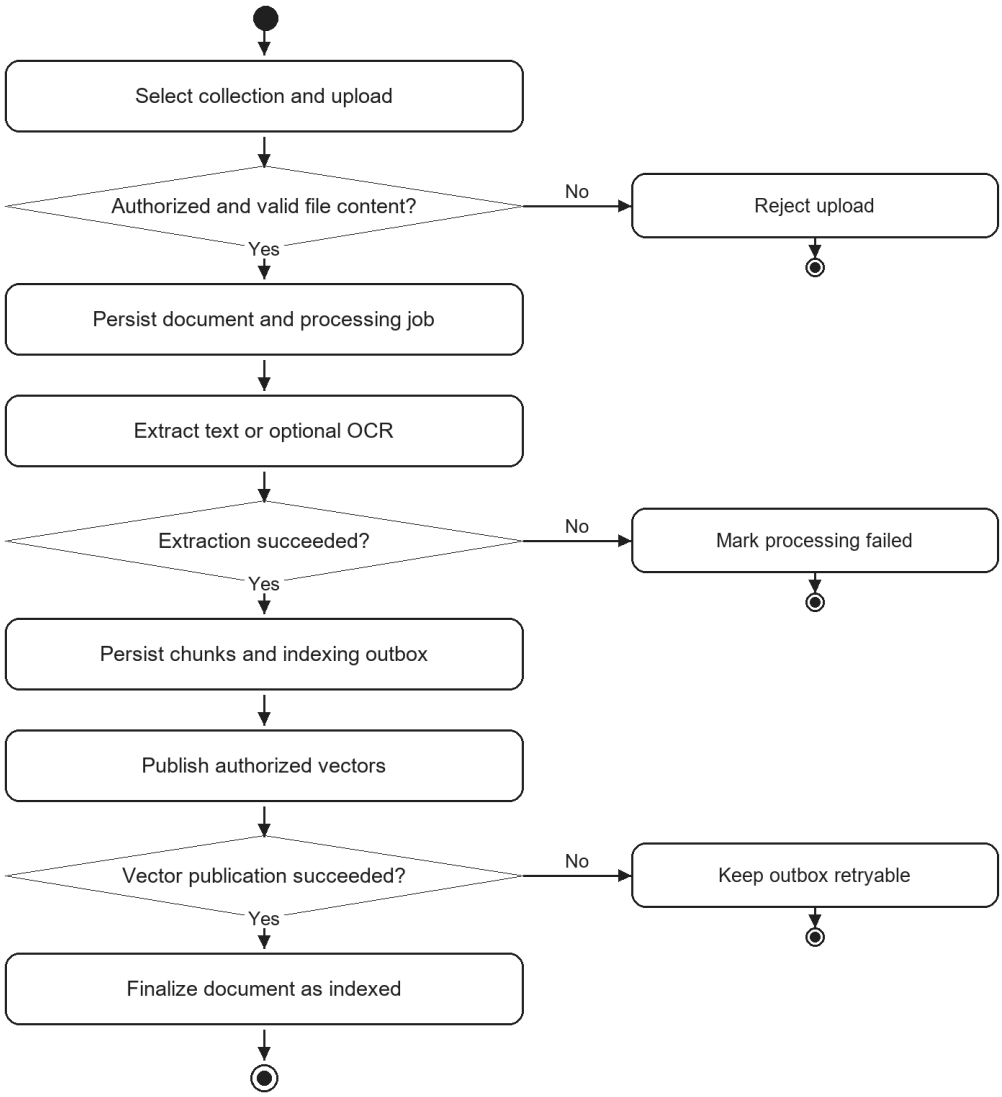


Figure 4. Document processing activity diagram

The diagram shows the main path. Extraction or indexing failure must record the failed state and its reason. A successful upload can still be awaiting processing. Pause is offered only during pausable stages.

Answer Generation Activity

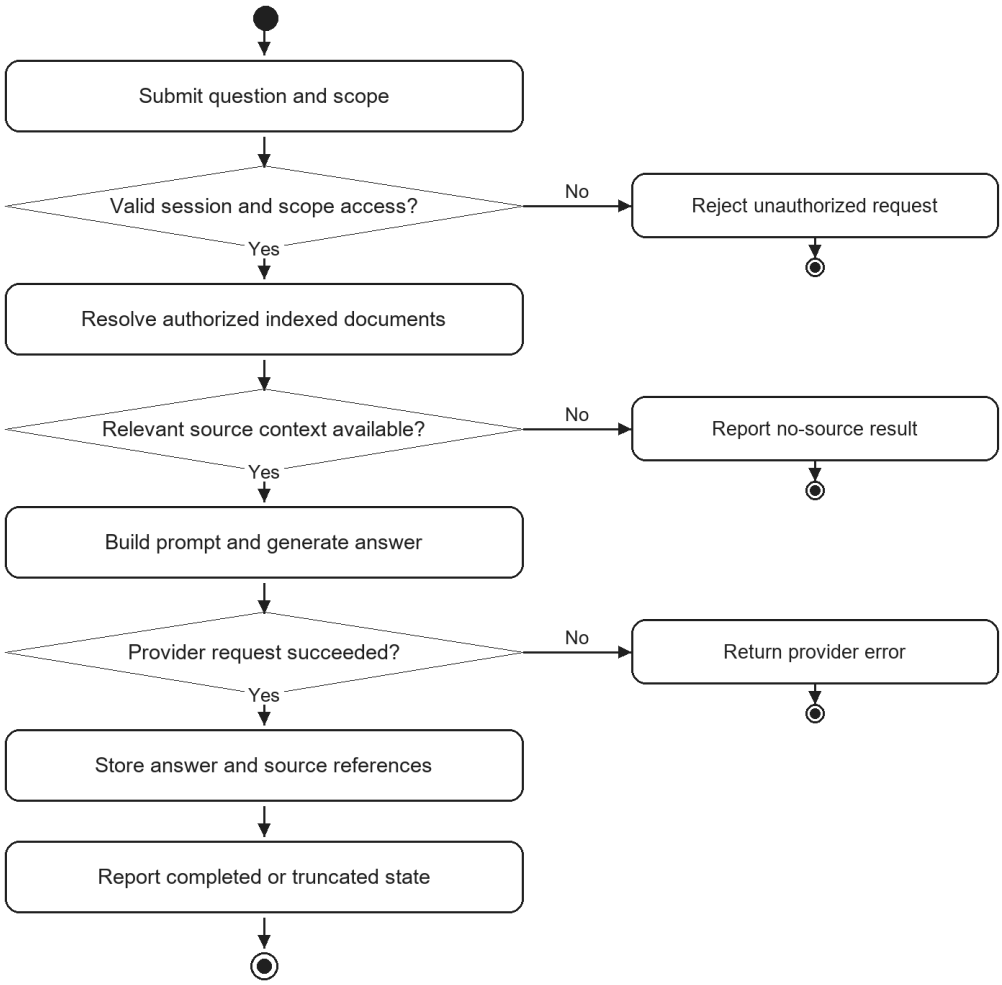


Figure 5. Answer generation activity diagram

The diagram shows source-grounded answering. If settings permit an answer without sources, it must identify its different basis. Reaching the generation length limit must set the truncated state.

Phase Three: Sign-in Sequence

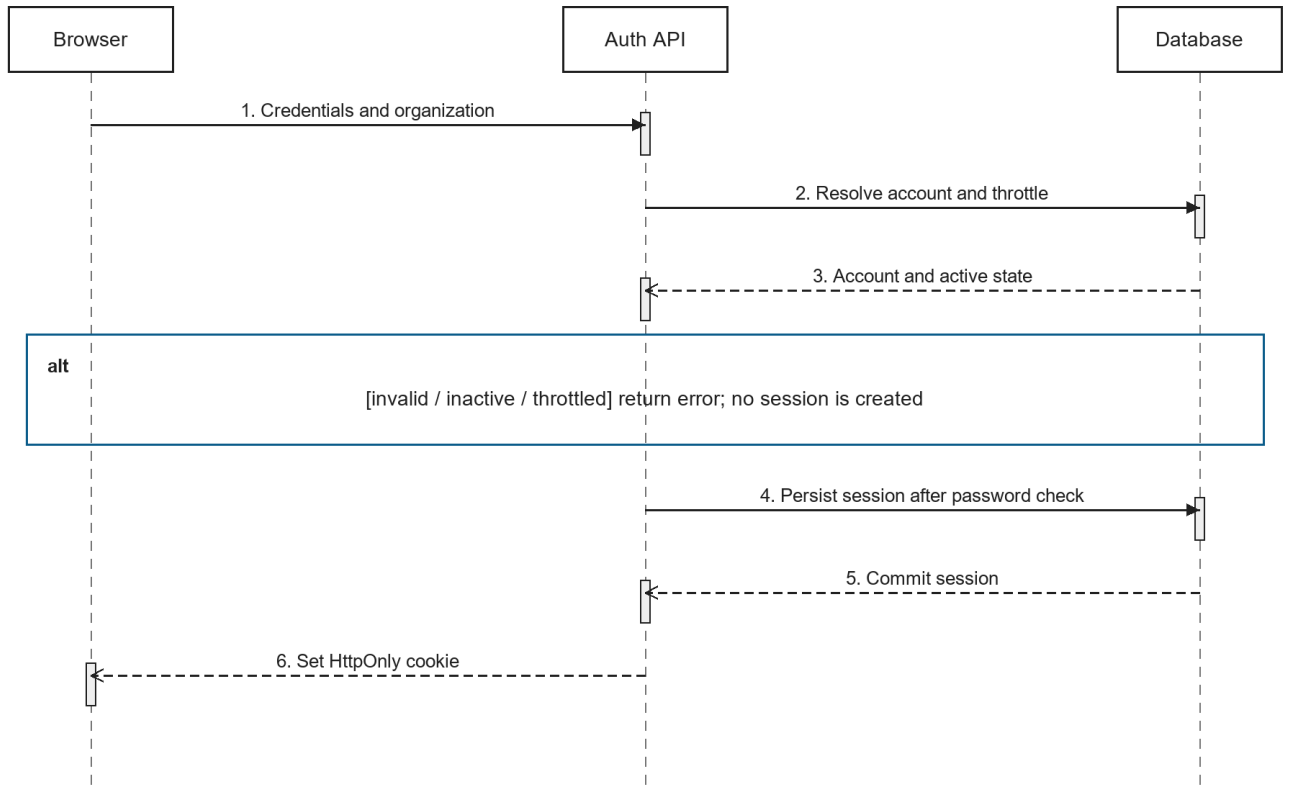


Figure 6. Successful sign-in sequence

The server verifies the password. The database stores no raw password. The invalid branch ends before session creation. Subsequent requests use the session cookie, and sign-out must revoke the associated session.

Upload and Indexing Sequence

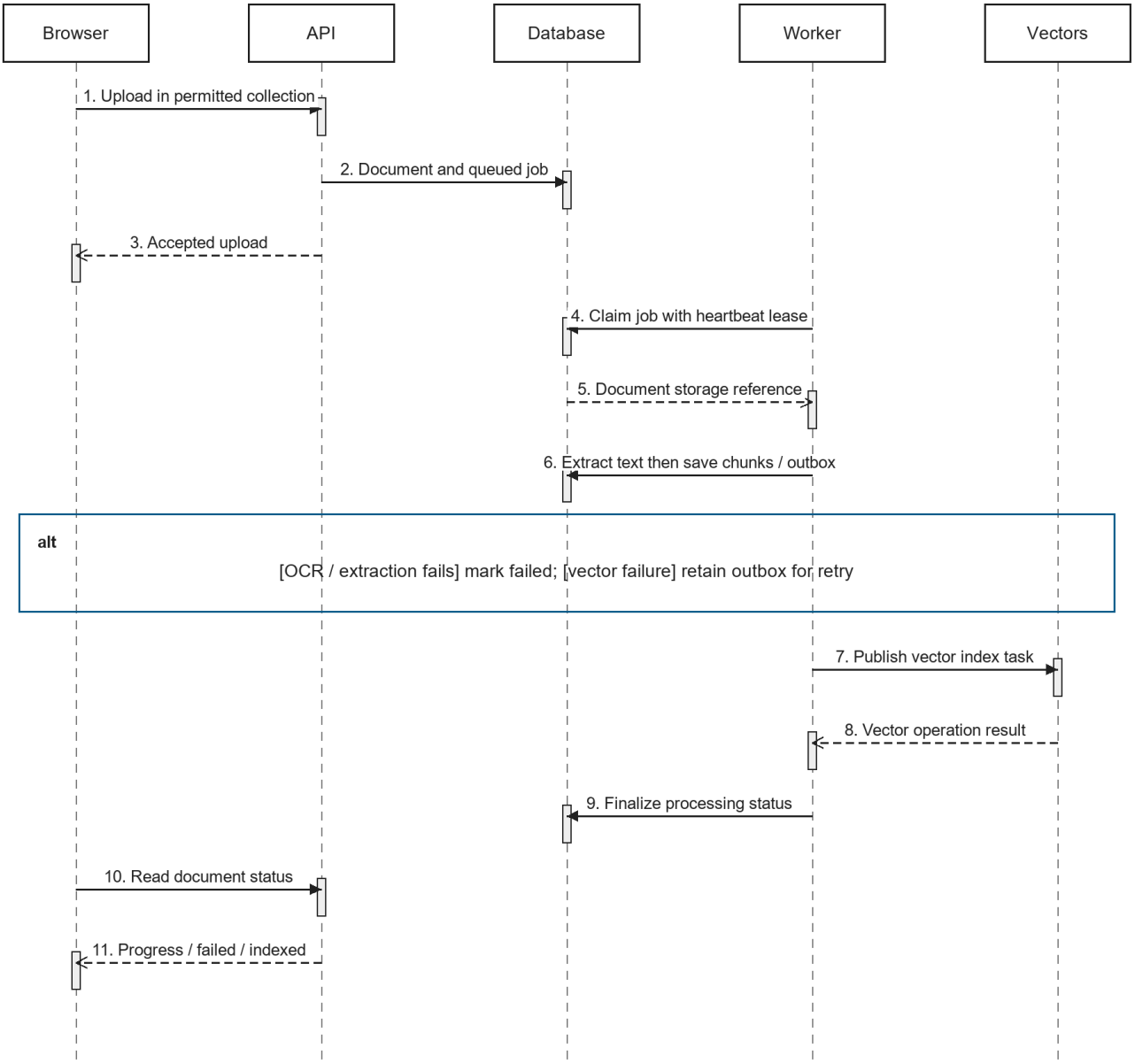


Figure 7. Asynchronous processing sequence

File reading, extraction, and OCR occur between claiming the job and storing chunks. They are omitted from the figure for legibility. SQL and Qdrant have separate transaction boundaries. The outbox makes incomplete publication traceable. The initial API response does not indicate that indexing is complete.

Question and Answer Sequence

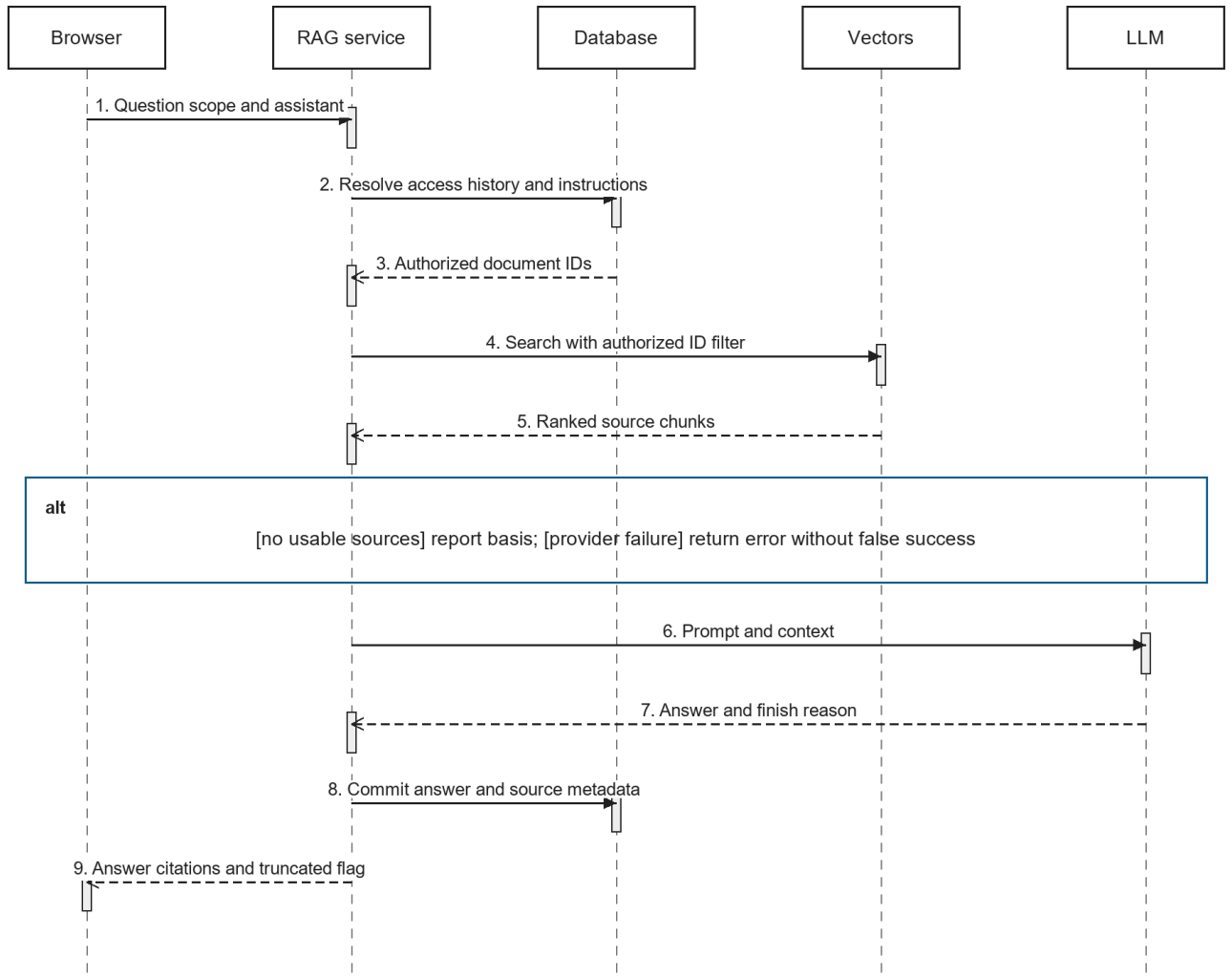


Figure 8. Source-grounded answer generation sequence

Access is checked before searching Qdrant. The model call is separate from the final write unit. Answer data belongs to persistent history only after successful storage. The API must map model and storage errors to suitable responses. Citations are built from the metadata of the relevant sources.

Identity and Knowledge Class Diagram

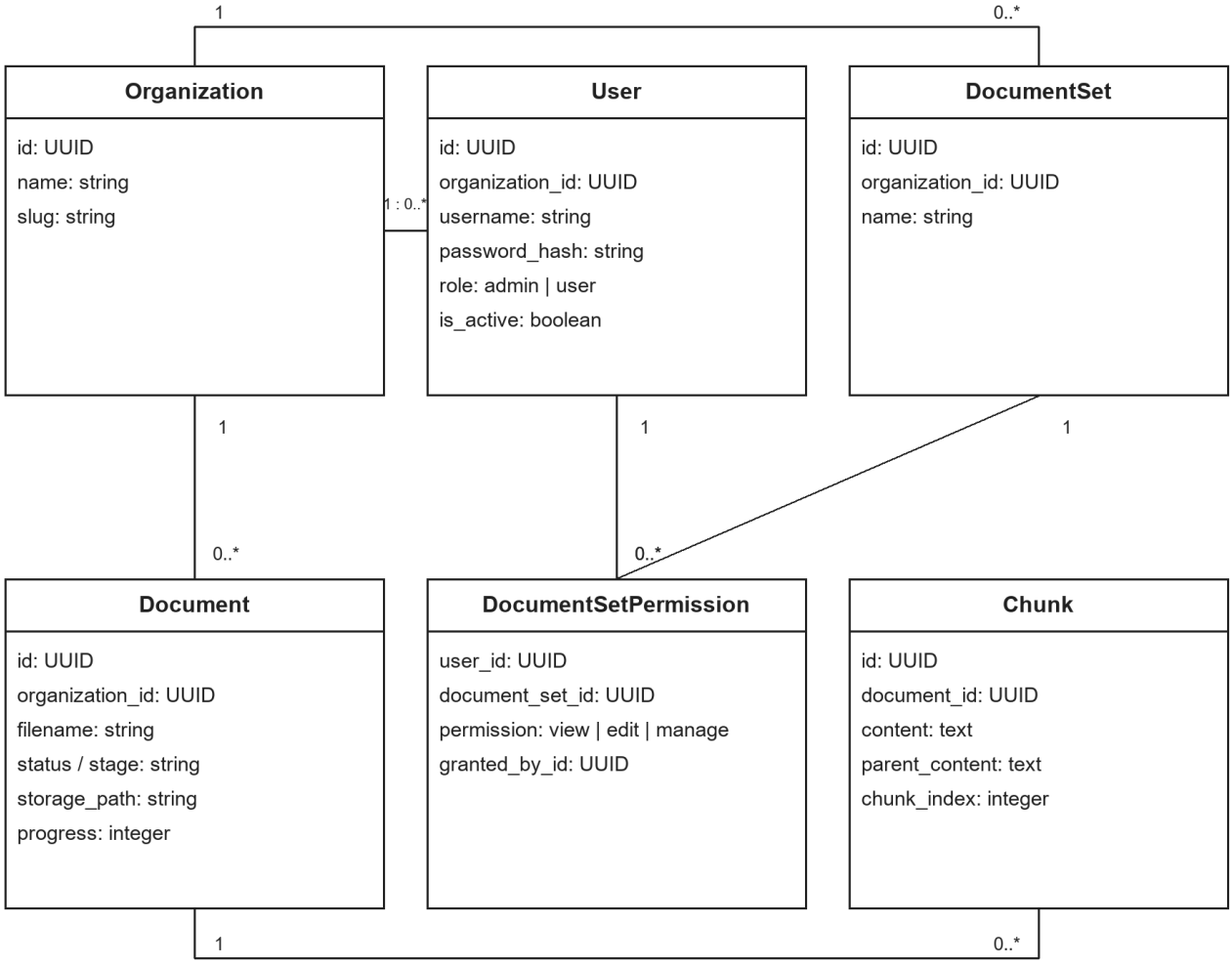


Figure 9. Identity and knowledge domain classes

The model is a simplified view of persisted entities. No invented methods have been added to the ORM classes. DocumentSet and Document have a many-to-many relationship through document_set_documents. That relationship and the set's organization association are omitted for legibility. Each permission record belongs to one user and one set, with the pair of IDs as its key. A document can have multiple chunks.

Conversation and Connector Class Diagram

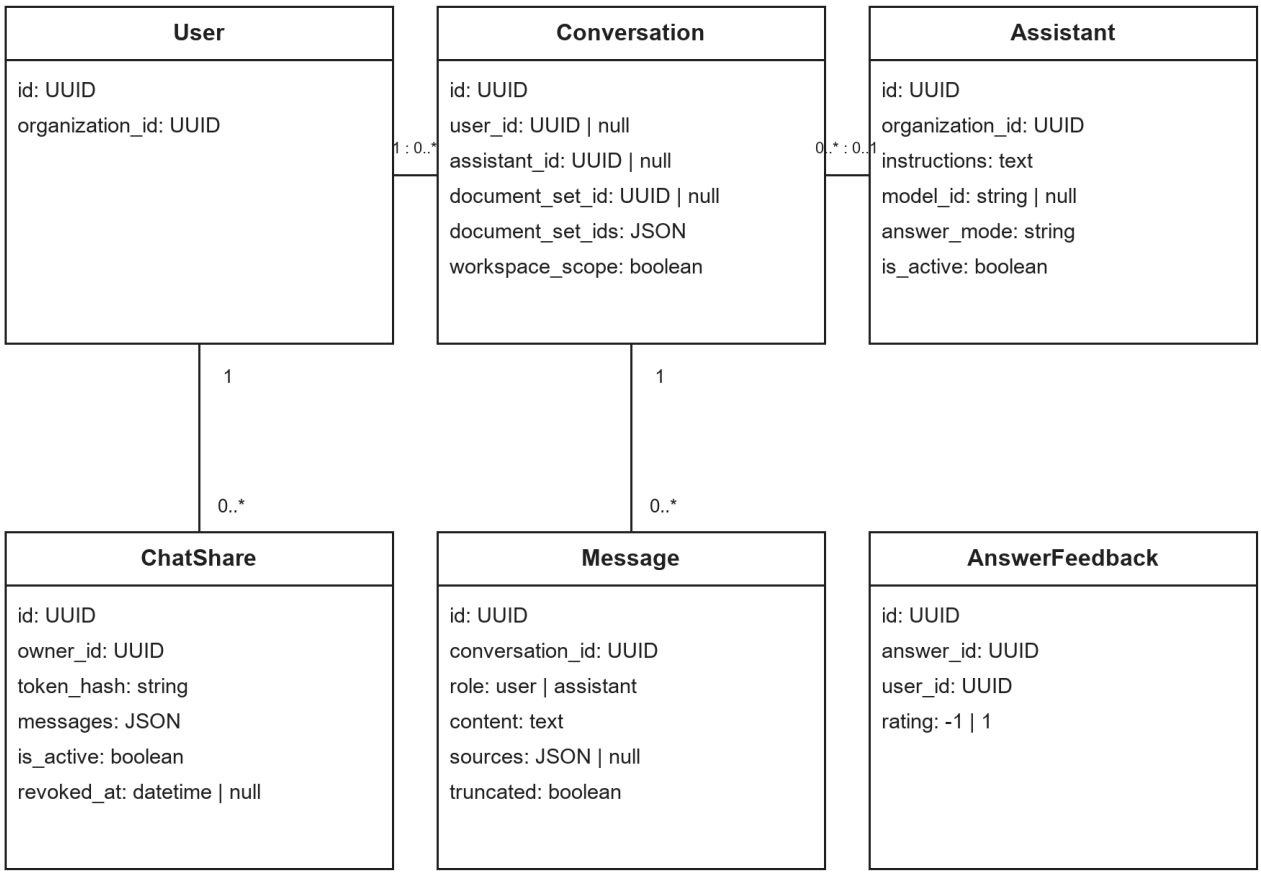


Figure 10. Conversation and connector domain classes

A conversation's association with an assistant is optional. An assistant connects to multiple knowledge sets through assistant_document_sets. Answer sources are recorded as JSON in Message. Each ConnectorItem refers to a connector and a document. Its external ID is unique within that connector.

Component Architecture and Responsibility Boundaries

Component	Responsibility	Technology
Web interface	Display pages, forms, processing status, and conversations	React, TypeScript, and Vite
API	Sessions, authorization, HTTP contracts, and error mapping	FastAPI
Application service	Coordinate flows and define write boundaries	Python
Repository	Query, insert, and update entities	SQLAlchemy
Database	Structured data and job status	PostgreSQL
Vector index	Retrieve chunks using authorization filters	Qdrant
Worker and Scheduler	Document processing and connector scheduling	Independent processes
Providers	Language model and optional OCR	OpenRouter and applicable adapters

The deployment includes the interface, API, document worker, scheduler, PostgreSQL, Qdrant, and shared file storage. Database rollback cannot undo every file or external provider operation. Consistency requires an outbox, retry, or appropriate compensating action.

Class diagrams depict the domain data model. Application operations such as ingest, sync, and answer belong to services. Repositories do not commit independently in this design. The service defines the write unit. This layer description does not claim that all remaining project paths follow it uniformly.

Requirements Traceability and Acceptance Scenarios

Requirement	Use case	Proposed acceptance test
FR-01 to FR-03	UC-01	TC-01 Valid and invalid sign-in, and session revocation after sign-out
FR-04 to FR-06	UC-04	TC-02 Reject a direct request to an unauthorized set.
FR-07 to FR-15	UC-02	TC-03 Process a valid file and independently reject an invalid one.
FR-16 to FR-21	UC-03 and UC-08	TC-04 Answer only from authorized scope and detect truncation.
FR-22 to FR-24	UC-05	TC-05 An assistant grants no new access to an unauthorized set.
FR-25 to FR-27	UC-06	TC-06 Repeated sync and lease contention do not cause inconsistent execution.
FR-28 to FR-30	UC-07	TC-07 Reject revoked links and unauthorized feedback.
FR-31 to FR-36	Administrative and UI flows	TC-08 Check role limits, inline feedback, and waiting states.

Disruption Testing

Test separately for worker interruption during extraction, Qdrant failure during outbox publication, model timeout, and commit failure. After an outage, document and message states must match the actual result. For permission changes, check the UI list, direct API requests, and vector retrieval.

Proposed Capabilities

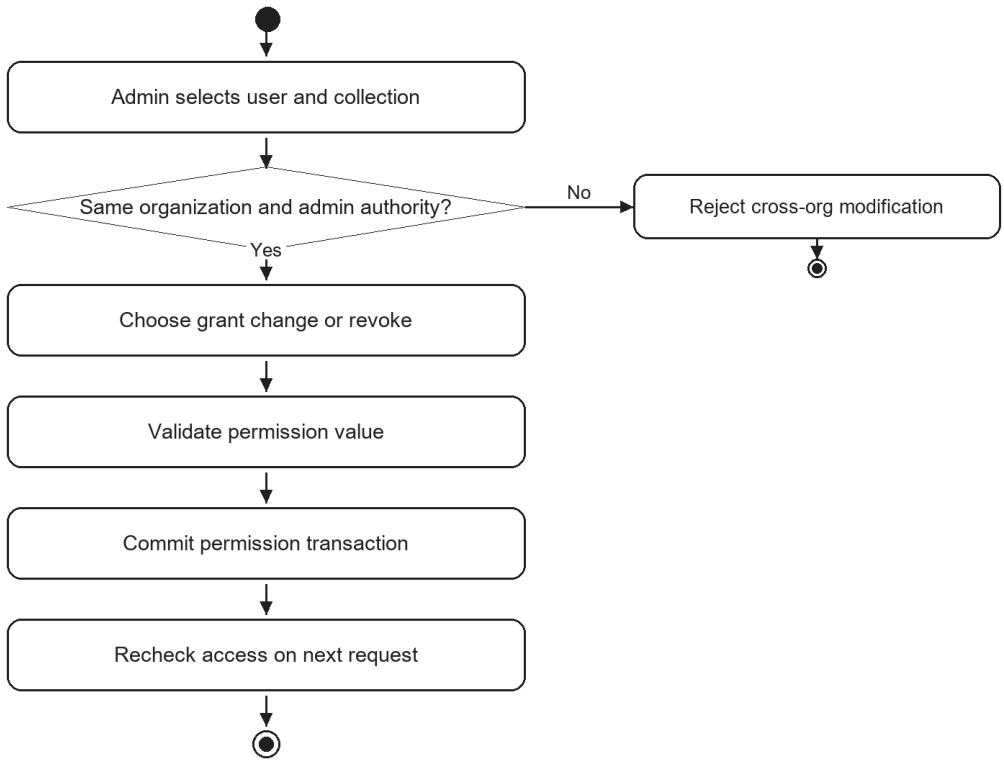
Independent Organization Sign-in and Management

To complete the multi-organization experience, the sign-in form should accept an organization identifier or map a dedicated organization domain to it. Creating an organization and initial administrator, managing organization status, and preventing disclosure of account existence in other organizations require separate design. The organization ID infrastructure exists. Full organization management is not reported as part of the current interface.

Personal Agent Integration

The current assistant has instructions and sources, and connectors import content. Integrating a personal agent requires a tool execution contract, request validation, organization authorization, time and cost limits, and operation logging. The current-version diagrams omit this capability to distinguish autonomous execution from RAG answering.

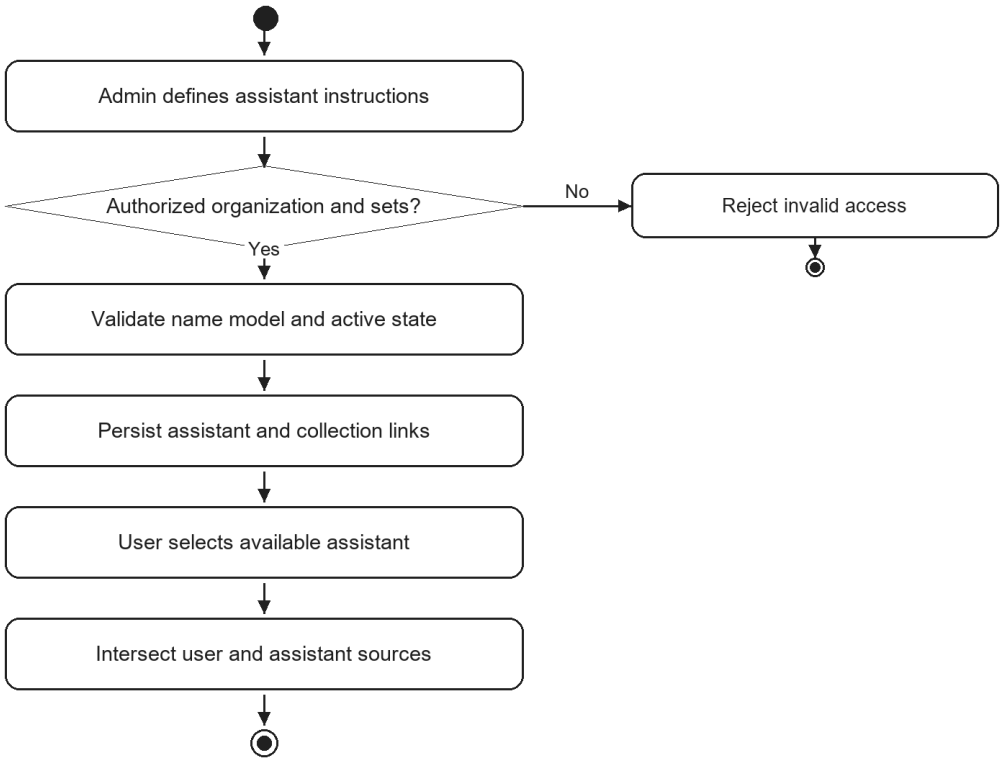
Access Assignment Activity



Access assignment activity diagram

The figure shows the main path, authority or status checks, and rejection branch. An error branch ending does not indicate success. The server performs all checks. Displaying controls in the interface does not replace authorization.

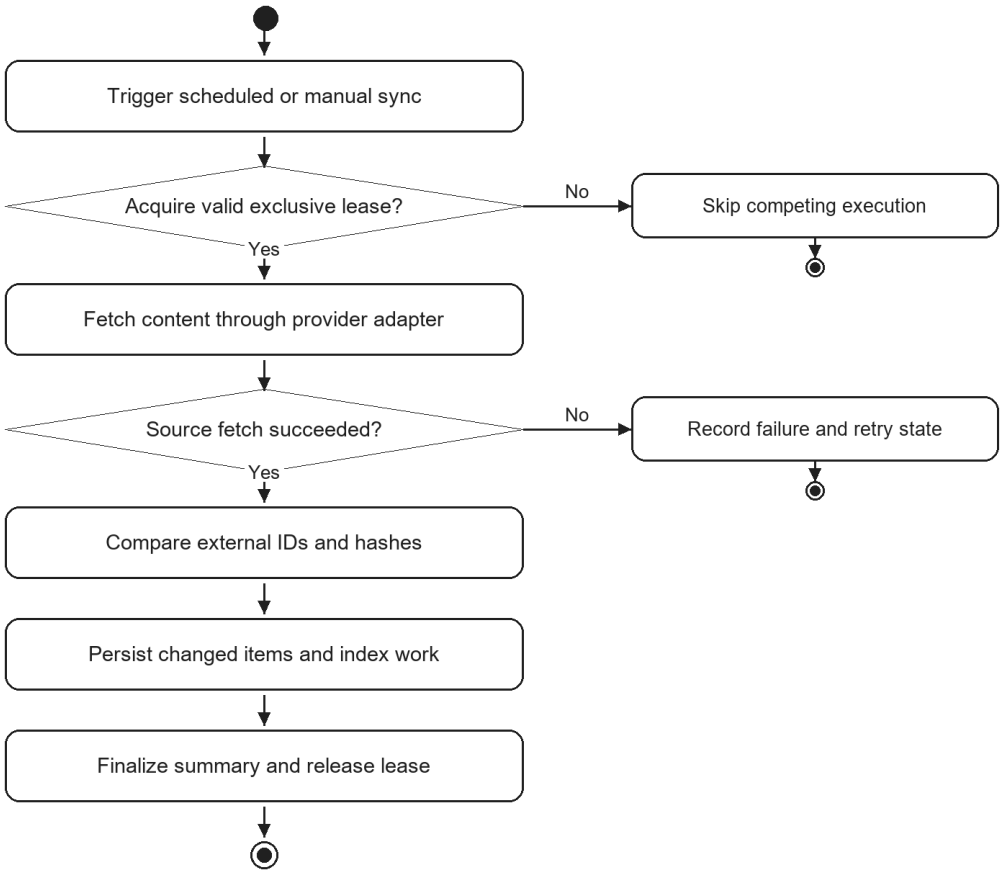
Assistant Definition Activity



Assistant definition activity diagram

The figure shows the main path, authority or status checks, and rejection branch. An error branch ending does not indicate success. The server performs all checks. Displaying controls in the interface does not replace authorization.

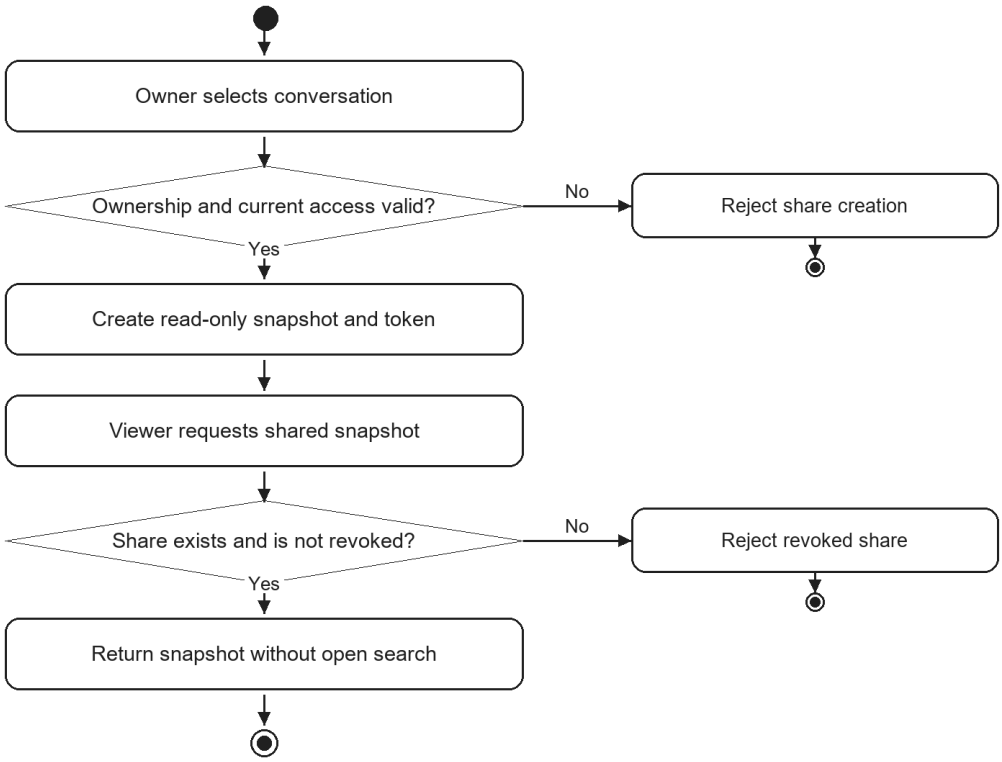
Connector Synchronization Activity



Connector synchronization activity diagram

The figure shows the main path, authority or status checks, and rejection branch. An error branch ending does not indicate success. The server performs all checks. Displaying controls in the interface does not replace authorization.

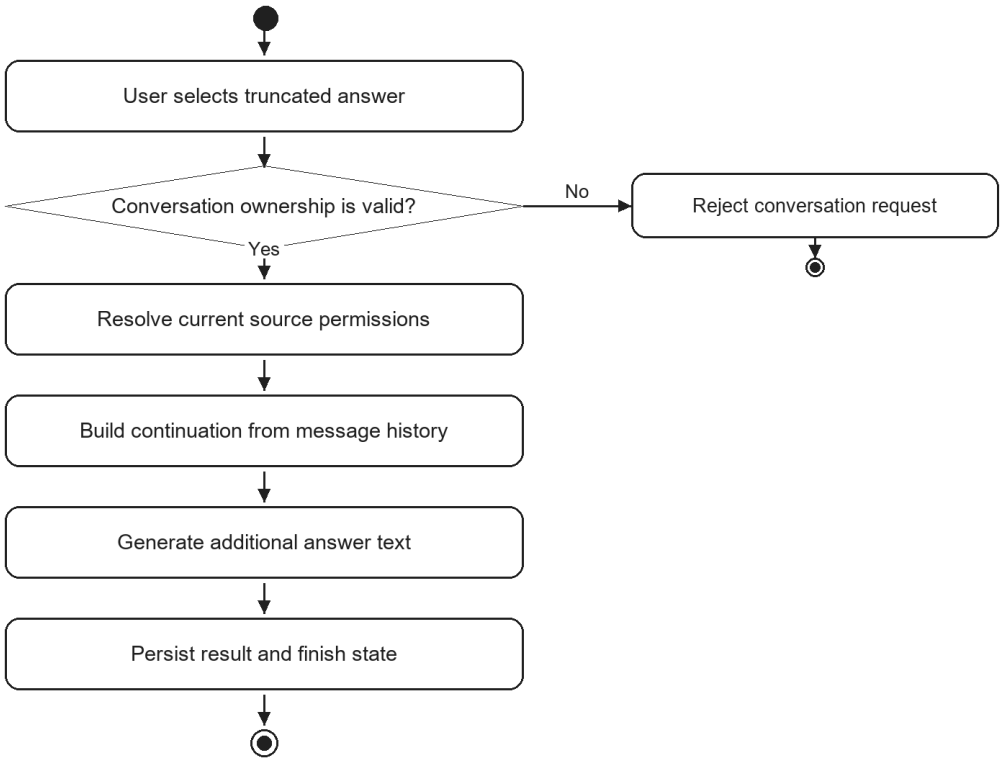
Conversation Sharing Activity



Conversation sharing activity diagram

The figure shows the main path, authority or status checks, and rejection branch. An error branch ending does not indicate success. The server performs all checks. Displaying controls in the interface does not replace authorization.

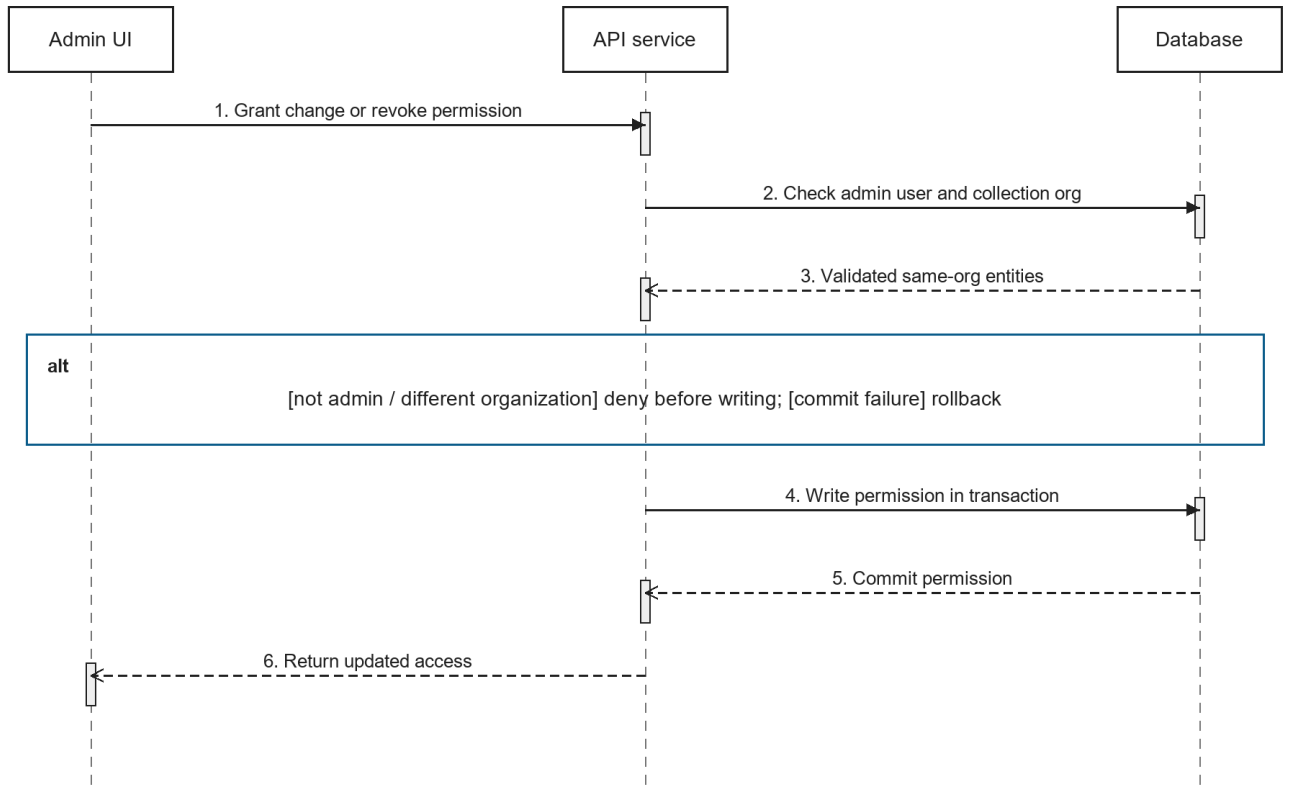
Answer Continuation Activity



Answer continuation activity diagram

The figure shows the main path, authority or status checks, and rejection branch. An error branch ending does not indicate success. The server performs all checks. Displaying controls in the interface does not replace authorization.

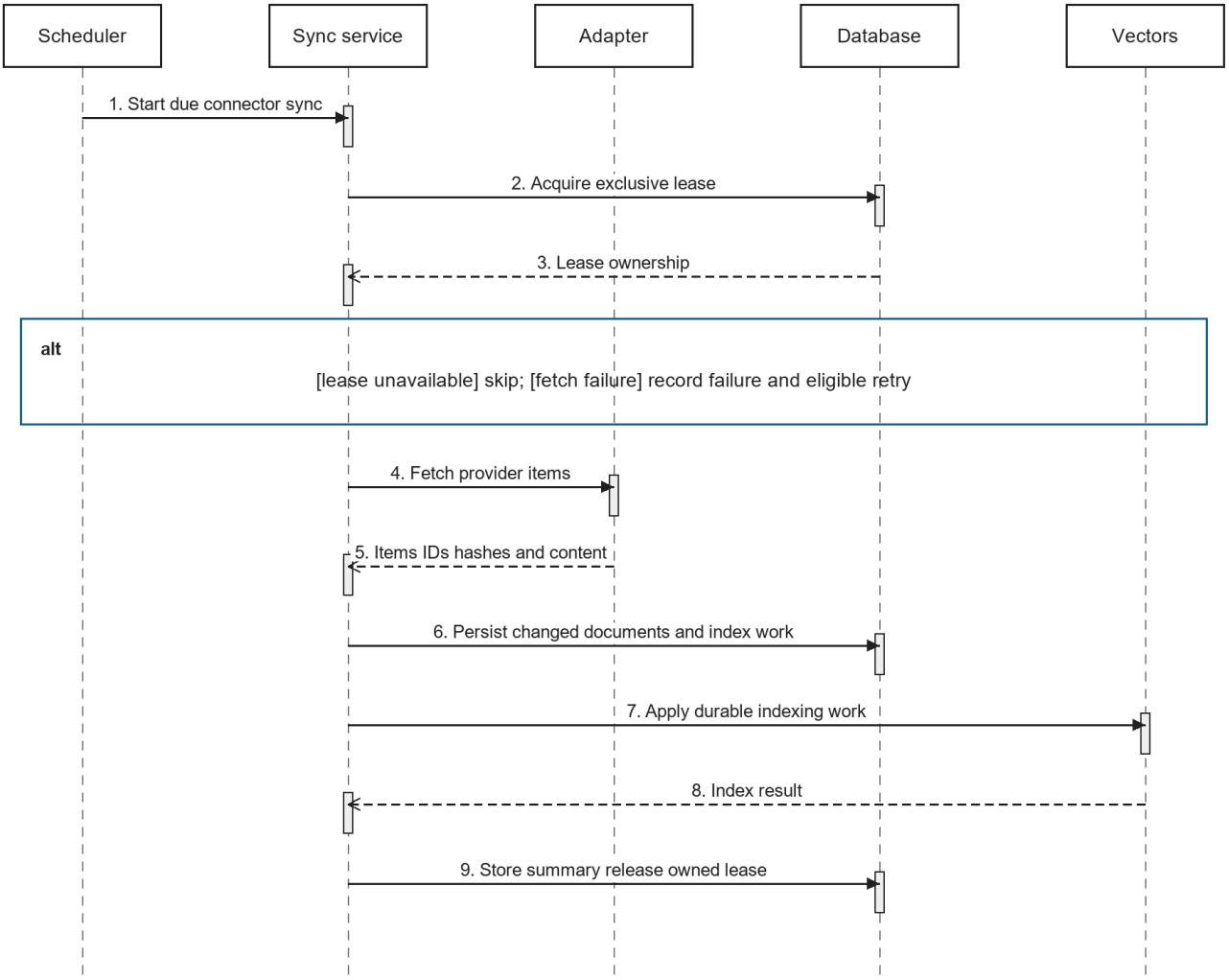
Permission Change Sequence



Permission change sequence diagram

Solid arrows show requests and dashed arrows show responses. The alt frame marks an alternative or error path. The recorded result must match a successful transaction. An external failure must not be reported as success.

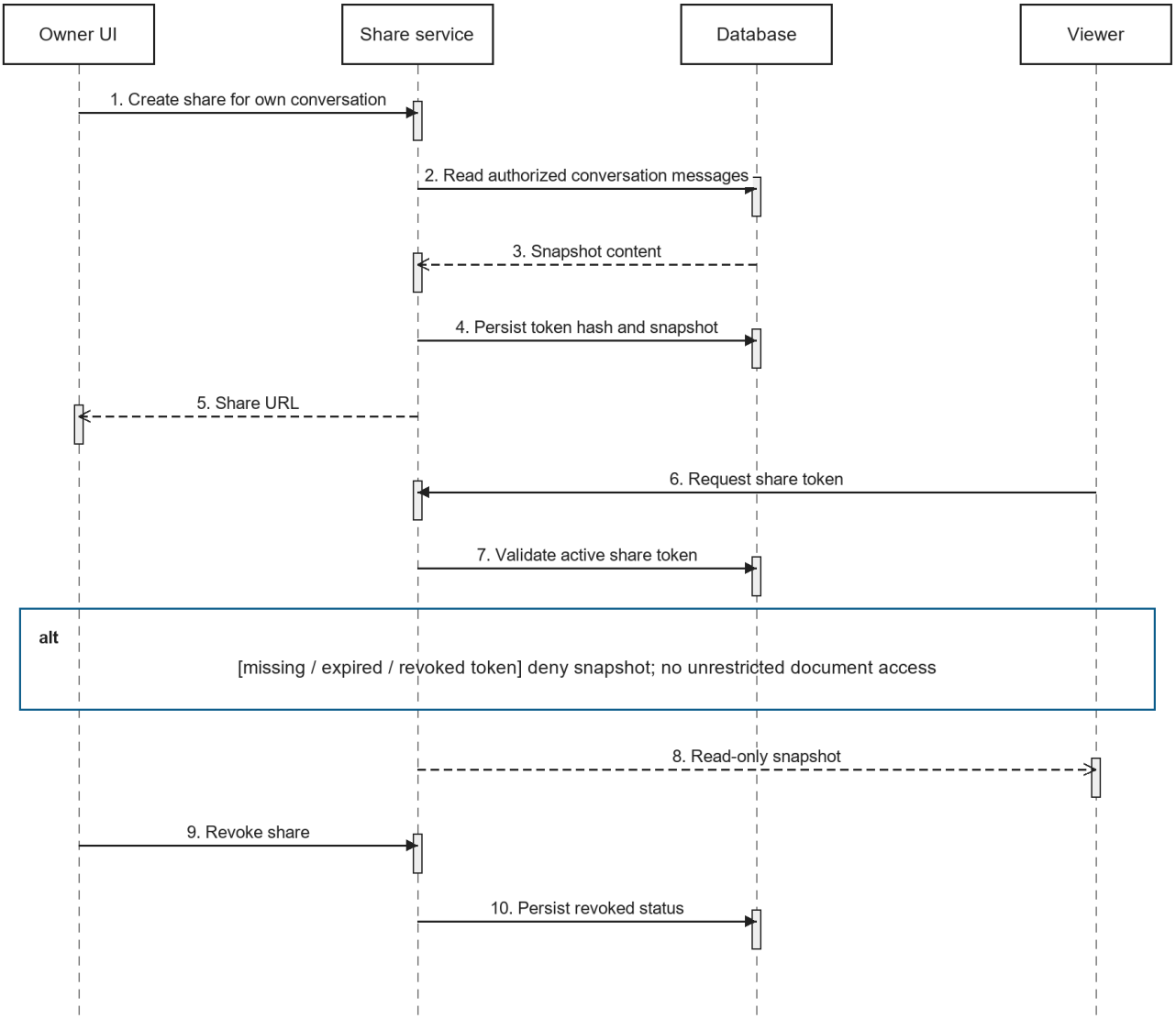
Source Connection Sequence



Source connection sequence diagram

Solid arrows show requests and dashed arrows show responses. The alt frame marks an alternative or error path. The recorded result must match a successful transaction. An external failure must not be reported as success.

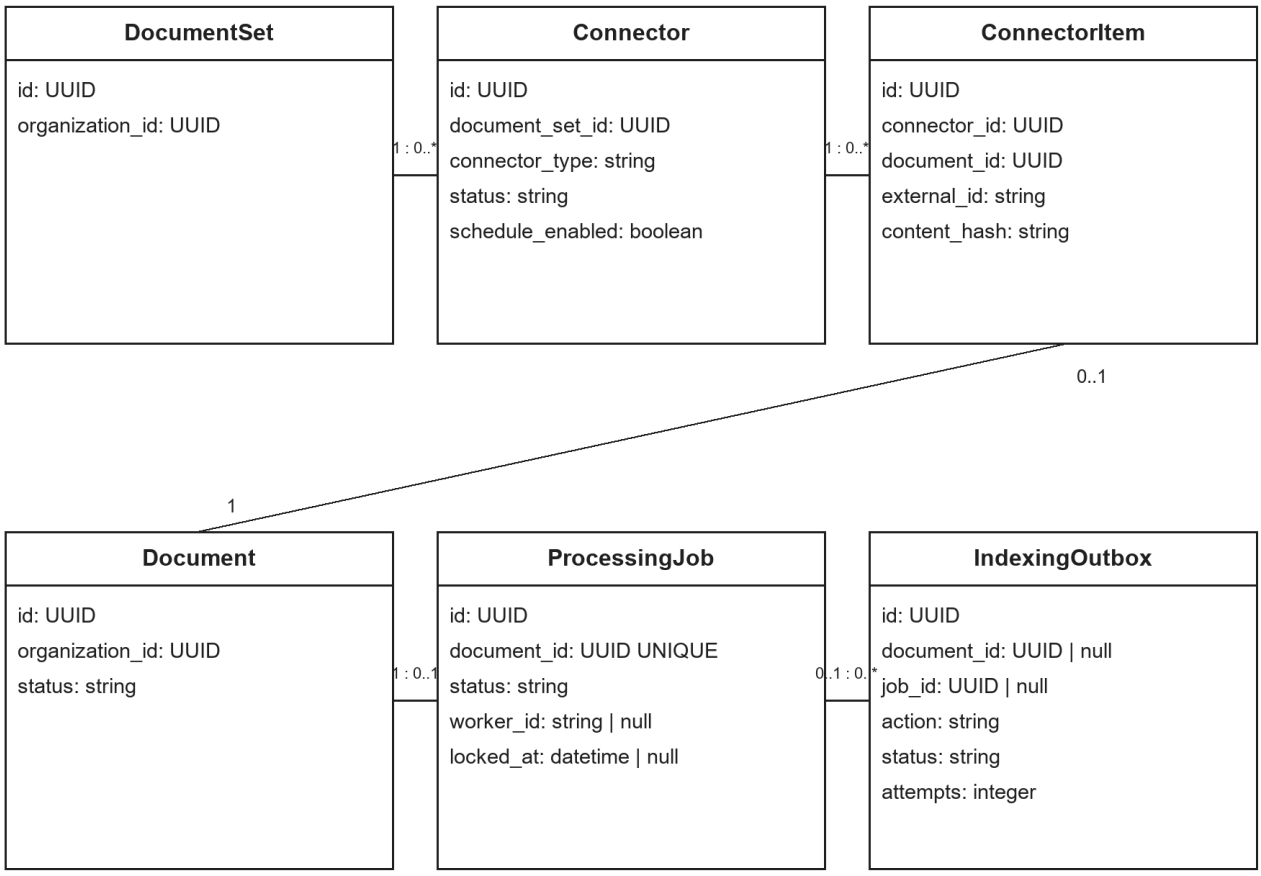
Sharing and Revocation Sequence



Sharing and revocation sequence diagram

Solid arrows show requests and dashed arrows show responses. The alt frame marks an alternative or error path. The recorded result must match a successful transaction. An external failure must not be reported as success.

Connector and Processing Classes



Connector and processing class diagram

Each connector belongs to one set and holds multiple external items. Each item refers to a document. The processing job and outbox are independent execution lifecycle models. The outbox does not share a transaction with Qdrant and is tracked by its own state.

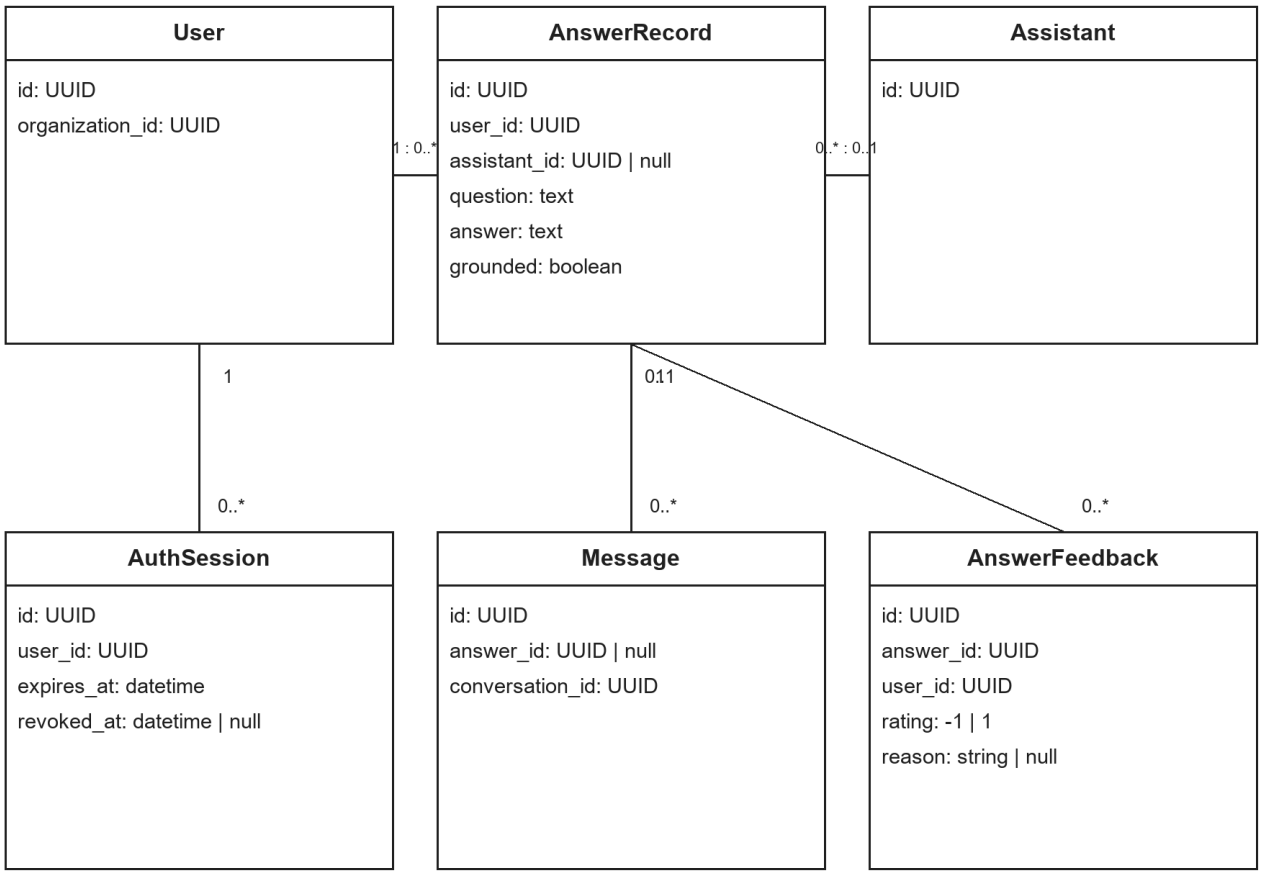
Many-to-Many Relationships for Knowledge and Assistants



Many-to-many relationships for knowledge and assistants

Join tables with composite keys resolve two many-to-many relationships into one-to-many relationships. Zero-to-many allows no association or multiple associations. Each join row belongs to exactly one entity on each side.

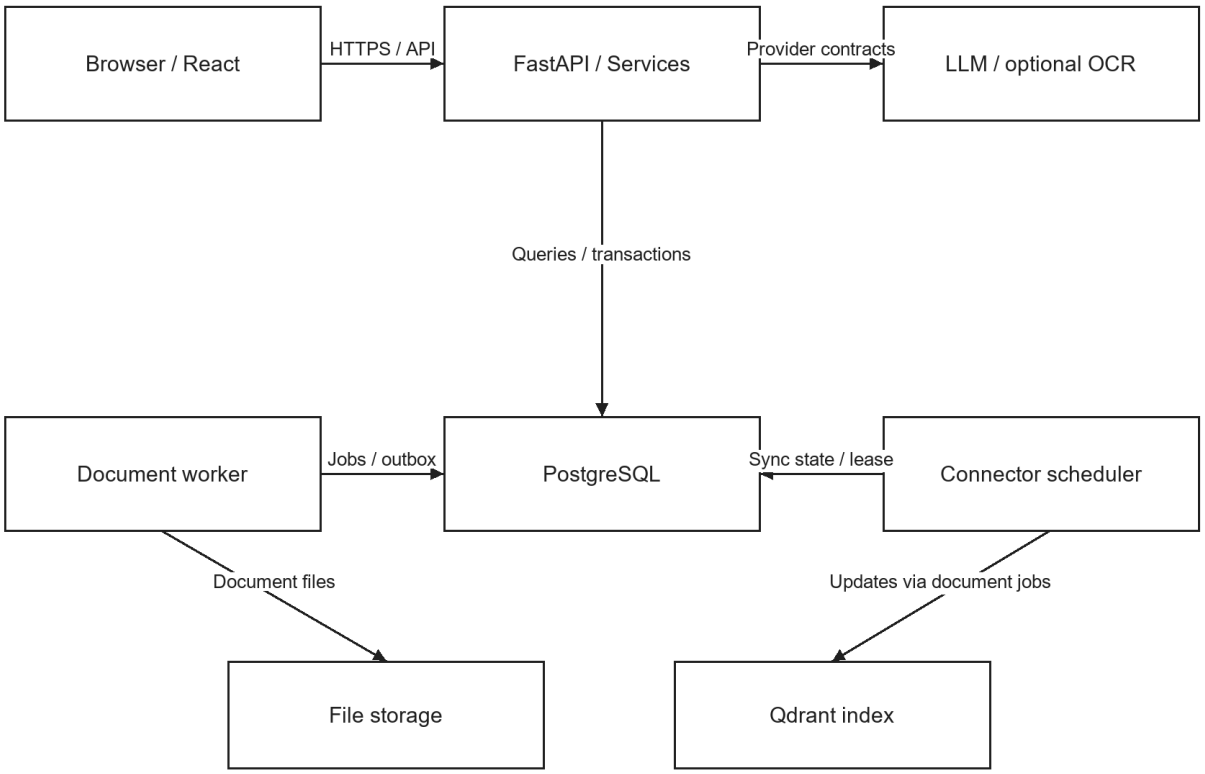
Session and Answer Evaluation Classes



Session and answer evaluation class diagram

Feedback references AnswerRecord rather than free-form message text. Message can reference an answer record. Sessions and conversation shares belong to users. ChatShare stores a JSON snapshot of messages and has no conversation_id foreign key. The association between the reviewing user and feedback is omitted for legibility.

Deployment and Component Dependencies



Runtime dependencies (direct or delegated through document jobs)

Process	SQL	Files	Vectors	LLM	OCR	Sources
API	yes	yes	yes	yes	optional	-
Document worker	yes	yes	yes	yes	optional	-
Connector scheduler	yes	via jobs	via jobs	via jobs	via jobs	yes

Deployment and component dependency diagram

The API, document worker, and scheduler run as separate processes. File storage, database, and index must be shared and accessible to the relevant components. Model and OCR services are outside the SQL transaction boundary. The figure shows each dependency's main flow for legibility.